

Research article

RPL*: An Explainable AI-based routing protocol for Internet of Mobile Things

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ABSTRACT

The Internet of Mobile Things (IoMT) is an emerging paradigm of Internet of Things (IoT) with special focus on enabling mobility to the ‘things’. Several IoMT applications such as group of robots or drones performing collaborative search and rescue operation, identification of mines, warehouse management, goods delivery, etc can be considered as examples of IoMT systems. In the applications mentioned above, the nodes may send the information in a multi-hop manner to the root or coordinator node which may be static or mobile. While the Routing Protocol for Low Power and Lossy Networks (RPL) is extensively utilized in static IoT networks, it encounters significant limitations in handling mobility and providing resilience against routing attacks in mobile IoT networks. In this work, we propose a modified RPL, RPL* which is robust to handling mobility in nodes and is resilient towards routing attacks. In RPL*, any deviation from the normal behaviors of the network are identified as anomalies using an unsupervised Explainable Artificial Intelligence (XAI) strategy. In RPL*, we propose a novel mobility detection mechanism that will identify the mobility in the network in an energy efficient manner without incurring additional communication overhead. To maintain the connectivity with parent node, we propose a novel proactive connectivity management mechanism in RPL* which will ensure a smooth transition from one parent to another if required, thus avoiding the network partitioning due to mobility. The performance analysis of the system has demonstrated an improvement in packet delivery ratio of the mobile nodes by 40% due to the proposed RPL* when compared to RPL. Also, the proposed XAI strategy provided an F1-score of over 95% for the detection of sink hole and black hole attacks in the tested IoMT network scenarios. It was observed that RPL* improves the performance of the IoMT network when compared to RPL. However it may be noted that the mechanisms introduced to support mobility does not lead to a drop in PDR or increase in control packet overhead for static networks. Hence, RPL* can be considered as an alternative to RPL for IoT as well as IoMT networks.

1. Introduction

The Internet of Mobile Things (IoMT) is a paradigm of Internet of Things (IoT) with special focus on enabling mobility to the ‘things’ [1]. IoMT specifically attempts to address the challenges associated with the introduction of mobility in ‘things’. The connected autonomous vehicles, robots, drones, unmanned aerial vehicles (UAV) etc., are examples of ‘things’ in the context of IoMT [2–5]. Routing among the ‘things’ is a very critical network function that affects the overall performance of an IoMT system [6,7].

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Some of the representative usecases that require routing among the ‘things’ in an IoMT system are given below. A group of soldiers navigating a tough mountainous terrain with multiple IoT sensor nodes mounted on the body and the data from the sensor nodes from all the soldiers in the group can be collected at the node worn by the person moving with the team and monitors the status of health of the soldiers [8]. Another application in the healthcare field includes the monitoring of assets or equipments within a hospital to ensure that the required asset can be accessed in the shortest time [9]. The position of all the assets can be tracked at a coordinator node in the network. Robots in a warehouse may collaboratively navigate, pick up and move objects from one place to another. To ensure co-operation among the robots and for ensuring the optimal completion of the tasks, the robots may be required to transmit the information to a coordinator robot or root node [4,5]. In this paper, we use the terms root or coordinator interchangeably to indicate a node at which the information from the network is collected in a converge cast manner. A group of robots may also collectively perform surveillance to uncover mines in a war zone or participate in goods delivery [10]. UAVs are proposed for use in civil, and military applications including surveillance, reconnaissance, terrain mapping, precision agriculture, and so on. UAVs can also serve as first responders which can generate valuable information about an area to ascertain if it is safe for the human responders to enter. Typical examples include the use of UAVs or drones to inspect buildings or areas affected by fire, poisonous gases or flood. A group of UAVs/drones can be used to carry medical supplies to the areas which are inaccessible by road in the event of a disaster such as a fire, flooding, earthquake, etc [3]. In the usecases mentioned above, the robots may send the information in a multi-hop manner to the root node which may be static or mobile. Maintaining a communication route to the root node is a requirement for the nodes in the network and most of the nodes can be battery operated.

The widely accepted routing protocols in IoT networks are Adhoc On-Demand Distance Vector Routing (AODV), Lightweight on-demand adhoc distance vector routing (LOADng) and Routing Protocol for Low Power and Lossy Networks (RPL) [11]. AODV or LOADng are reactive or on-demand routing protocols in which route discovery and maintenance of the route is performed only when any information has to be transmitted. The route maintenance is a time consuming process and hence reactive protocols are not be suitable for mobile networks for ensuring the timely delivery of messages. RPL is a proactive routing protocol and is suitable for applications that are required to maintain multi-hop routes from the source node to the root node. RPL is maintained by Routing Over Low power and Lossy network (ROLL) group of Internet Engineering Task Force and is designed for resource constrained low-power and lossy networks [12]. RPL constructs a Destination Oriented Direction Acyclic Graph (DODAG) to organize the network topology [12]. RPL supports Point to Point (P2P), Point to Multi-point (P2MP), and Multi-point to Point (MP2P) communication. However, RPL is primarily designed for static applications. The major modifications required in RPL for its utilization as a routing protocol for an IoMT network are as follows.

- **Mobility discovery:** Mechanisms for detecting the mobility of nodes, taking proactive measures to prevent disconnection from the network and maximizing the packet delivery ratio at the root node are required. The mechanisms should be incorporated with minimal communication overhead.
- **Efficient Route Maintenance:** The mobility required to adapt to real-time situations in the IoMT applications may lead to changes in the network topology. The frequent updation and maintenance of routes as quickly as possible is required to prevent data loss. Due to limited onboard resources, the route updation should be done with minimal communication overhead in an energy-efficient manner.
- **Resilience towards routing attacks:** RPL networks are susceptible to routing attacks which may adversely affect its mission. Mechanisms are required to ensure resilience of RPL towards such attacks.

Certain prior works are reported in literature that are targeted towards the improvement of the performance of RPL protocol under mobility [13,14]. We use the terms nodes and motes interchangeability in this paper. As per our knowledge, the existing works in literature are tested only for scenarios in which the mobile nodes are not involved in the routing and forwarding of data. The effect of mobility on motes/nodes which can route and forward the data is very critical for use of RPL in IoMT networks and hence needs to be evaluated. Also, the detection of mobility in the network without additional message transmission among the nodes, reduction in the network disconnection time and making the RPL robust towards the routing attacks is required.

The main contribution of this research work are as follows.

1. In this work, we propose a modified version of RPL, RPL* designed to effectively handle node mobility while enhancing resilience against routing attacks. In RPL*, deviations from the normal behaviors of the network are identified as anomalies using an unsupervised Explainable Artificial Intelligence (XAI) strategy. This approach enables the detection of routing attacks as anomalies, subsequently facilitating the isolation of affected nodes and prompt network maintenance.
2. In RPL*, each node implements a novel mobility detection mechanism that will identify the mobility in its neighborhood in an energy efficient manner without incurring additional communication overhead. The parent selection process in RPL* will assign lesser priority to the mobile node or the node which experience mobility in its neighborhood compared to static nodes during parent selection. By doing so, it mitigates frequent link disconnections and reduces packet delivery losses, thus minimizing the likelihood of a mobile node becoming a parent.
3. To maintain the connectivity with parent node, we propose a novel proactive mobility management mechanism in RPL*. This mechanism ensures seamless transitions between parent nodes as needed, thereby preventing network partitioning due to mobility events.

The paper is organized as follows. Section 2 presents the background, related works and highlight the gaps in the research work. Section 3 elaborates the system model and the proposed routing protocol, RPL*. Section 4 presents the performance evaluation of the RPL*. Section 5 provides the conclusion and future work.

2. Background

RPL is a proactive distance vector routing protocol that constructs a Directed Acyclic Graph (DAG) with one or more Destination-Oriented Directed Acyclic Graphs (DODAGs) [12]. Information regarding DODAGs is transmitted across the nodes in the network through ICMPv6 control messages, namely DODAG Information Object (DIO), DODAG Destination Advertisement Object (DAO), DODAG Information Solicitation (DIS) and Destination Advertisement Object Acknowledgment (DAO-ACK). The initiation of the DODAG formation begins with the root node broadcasting a DIO message to its neighboring nodes. DIO message includes the essential information regarding the transmitting node which will be utilized by the receiving node for joining the DODAG. The nodes that receive the DIO messages, includes the transmitter of the DIO message into its parent list. The receiving node then computes its rank by utilizing the information in the DIO messages received by it and selects a preferred parent from the parent list. A node which has selected the preferred parent transmits a unicast DAO message to the preferred parent if it wishes to join the DODAG. The node that receive the DAO message provides a DAO acknowledgment if it accepts the transmitter node as a child node. This iterative process continues until all nodes are incorporated into the DODAG.

In RPL, the DIO messages are transmitted by the nodes on the expiry of the trickle timer. RFC 6206 provides a detailed overview of the setting of the trickle timer duration [15]. To reduce the overhead caused by the transmission of the repeated DIO messages, the duration of the tickle timer (trickle period) is increased exponentially by the node if no topological changes in the network are observed by it i.e., the network is in a stable state or is static. However, due to this, if a node moves away from the current position or leaves the DODAG, it will take a longer duration for the neighbor nodes to observe the non-receipt of DIO from the parent and this will delay the route maintenance process, leading to route breakages and packet loss. Also, the trickle period of the node is reset to the trickle timer's minimum duration on the detection of any inconsistency in the network, which results in a large communication overhead due to frequent transmission of DIO messages. In RPL, if a node does not receive a DIO message within a configurable duration, it can transmit a DIS message to prompt the neighboring nodes to respond with a DIO message [13]. Several researchers and practitioners have identified the need for improvement in the RPL protocol to make it robust towards the routing attacks and to support mobility in the nodes. The prominent works in literature that have addressed the mobility in RPL are discussed in this section. To reduce the drop in the packet delivery ratio of the network due to mobility, it is important to identify the impact of mobility on the nodes as quickly as possible and to take proactive network management operations to minimize the network disconnection time.

To detect mobility in RPL network, researchers have proposed changes to the implementation of the trickle timer and utilization of mechanisms such as Software-Defined-Network (SDN), etc.

Cobârzan et al. proposed the activation of a reverse trickle timer on the announcement of the mobility by a node through the DAO messages [16]. When a mobile node connects to the preferred parent, initially the interval between DIOs (tickle period) is set to a large value ' I_{max} ' under the assumption that it is likely to remain connected to the parent for a long time. After each DIO transmission, the trickle timer value is reduced by half and the process is repeated till the trickle timer is above ' I_{min} '. The trickle timer interval is adjusted in inverse to the operation of trickle timer in RPL in which the value of trickle timer interval is increased with time till a maximum value ' I_{max} ' is reached. This ensures quick identification of mobility and route updation in the network. However, in this protocol, it is required that the nodes announce their mobility through DAO messages, which increases the number of control messages in the network.

Mohammadsalehi et al. assumes that all nodes are aware of their velocity and position in the network using an Inertial Measurement Unit (IMU) and/or via Global Positioning System (GPS) [17]. Each node then calculates a metric, i.e., Time-To-Reside (TTR) from their neighbor nodes and selects the node with the longest TTR as its parent as the node with higher TTR may remain connected with the parent for a longer time. However, since the nodes position and velocity information has to be transmitted by nodes, the method is not energy efficient for use in battery enabled mobile networks.

In SDNMob, the mobile nodes are equipped with IMU sensors and the deployment requires an SDN controller which is aware of the precise location of the static nodes [18]. Each mobile node broadcasts a beacon, which contains its position and velocity information obtained using the IMU sensors. The static nodes which receive the beacon, append the measured Received Signal Strength Indicator (RSSI) and transmits it towards the SDN controller. The SDN controller determines the best static node as the parent node for the mobile node. However, this is not a communication efficient method and requires introduction of SDN controller in the network to select the parent nodes. From the literature review, it was understood that there is a need for introducing a mechanism to identify mobility in the network without incurring additional hardware or communication overhead in RPL.

In RPL, the nodes are assigned rank based on the objective functions, Objective Function Zero (OF0) and Minimum Rank with Hysteresis Objective Function (MRHOF). The metric, Expected number of Transmissions (ETX) is taken as the MHROF and the minimum hop count is used as the OF0. ETX is ratio of successful packet delivery in the downward and upward directions in the network. The nodes are ranked based on the minimum hop count to the root node and the ETX. The node with the lowest rank is selected as the best parent. To account for mobility, researchers have proposed the use of RSSI as an important factor for parent selection. Ko et al. suggested a MoMoRo fuzzy estimator to detect the link quality between mobile and static nodes [19]. When mobility is detected, the ETX, RSSI, Standard Error of Regression (SER) information are computed from the nearby neighbors and the quality of the links is then estimated with a fuzzy estimator. The node which provides the most robust link is selected as the parent. In the Fuzzy based Mobile Objective Function (FMOF) technique, the authors have utilized ETX, average RSSI and hop count to the root node to select the best parent for the mobile node in case if the situation necessitates fast parent switching due to mobility, thus reducing the hand-off delay [20].

Another major challenge in RPL is the mobility management, after the discovery of mobility in the network. In mRPL, the authors have proposed a smart handoff mechanism to maintain the connectivity between nodes in a dynamic environment [21,22]. In mRPL, the mobile nodes monitor the channel and if they do not receive any DIO packet within a given period of time or if the average RSSI of the received packet is less than a threshold, a handover process is initiated by broadcasting a burst of DIS. The neighbor nodes that receive the DIS burst measure the average RSSI and transmit the average RSSI in a reply message. The mobile nodes select the node with the best average RSSI as the preferred parent. mRPL⁺ introduces the advantages of both hard and soft handoffs, leading to improvements in packet delivery ratio and reduction in the handoff delay [22]. mRPL implements hard handoff in which a new parent discovery and selection is carried out only after the detachment from the previous parent. In the soft handoff, the mobile node drops the connection with the parent only after ensuring a stable connection with the new parent. In this mRPL⁺, the authors have assumed that the mobile nodes are not involved in forwarding or routing.

In Energy-efficient and Mobility-Aware Routing Protocol (EMA-RPL), to reduce the burden on mobile nodes to detect the new parent, the static nodes detect the mobility of mobile nodes based on RSSI change and triggers the mobile nodes to broadcast the burst DIS message [23]. Neighbor nodes measure the average RSSI values and append them into the DIO reply message. The static nodes that detect mobility, first perform the comparison of the RSSI and select the best static parent and informs the best parent for each mobile node. It may be noted that the static nodes were used as routing and forwarding nodes in the studies mentioned above, and mobile nodes were only placed as leaf nodes.

The RPL protocol lacks the capability to detect attacks or anomalies and recover from them. Notable works, in the beginning of 21st century, introduced the term 'resilience' in the context of engineering design [24,25]. The term 'resilience' can be defined as "the ability to recover quickly from change" [24,25]. Al-Shammari et al. emphasized the importance of incorporating resilience through effective routing strategies in IoT networks and identified that proactive routing strategies can be introduced to resilience in the IoT systems [26]. C. Berger et al. have elaborated the term resilience by identifying that resilience can be introduced into the system by adding security or dependability features into the system, which can be implemented through behaviors such as absorption, recovery, adaptation, resistance while facing changes such as attacks, disruptions, accidental faults or planned changes [27]. Heurtefeux et al. have identified that resilience can be introduced into the RPL system through changes in the objective function of RPL leading to new parent selection under adverse conditions. The authors have also considered the transmission of duplicate messages in the network to cater to message losses in the network. Mobility in IoMT makes the identification of routing attacks as it is difficult to distinguish if the packet losses are due to mobility and routing attacks. Developing resiliency towards routing attacks is an important requirement in IoMT systems.

SM-Trust suggests modifications in RPL to support both mobility and routing attacks [28]. The authors proposed a preferred parent selection algorithm in which they calculate the trust index of nodes based on the following metrics: energy level, direct trust, recommended trust by one hop neighbor, RSSI, mobility status, and success rate of packets forwarded by a node using fuzzy logic. The parent is selected based on the calculated trust. In order to calculate the direct trust, the nodes must overhear other nodes, which increases overhead and energy consumption. Also, the recommended trust can be corrupted due to the bad mouthing attacks, in which the malicious nodes in the network share false information about the nearby nodes.

Yavuz et al. have simulated three types of attacks (rank decrease, hello flooding, version number attack) in RPL network under static scenario [29]. The authors extracted features such as transmission rate, reception rate, transmission total time, reception total time, transmission average time, reception average time, source count, destination count, DIS/DAO/DIO message count and using an Artificial Neural Network (ANN) achieved an accuracy of approximately 98% in the identification of the type of attack [29]. Medjek et al. used Decision Tree (DT) and Random Forest (RF) techniques to identify the type of attacks in static RPL network [30]. The authors have used the same features as used by Yavuz et al. and achieved 99% accuracy in the detection of sinkhole, blackhole, decrease rank, selective forwarding and version number attack. Kamaldeep et al. have simulated various attacks (i.e. UDP flooding, ICMPv6 flooding, selective forwarding, blackhole attacks, and DIS flooding attacks) that are generated at various protocol layers [31]. They used the same features and topology for data generation as Yavuz et al. and achieved an accuracy in classification of approximately 99% using RF classifier.

Mohammed et al. utilized techniques such as KNN, Support Vector Machine, Naive Bayes, Multi-layer perceptron (MLP) for attack detection on the IoT Routing Attacks Dataset (IRAD) [17]. The authors obtained best performance using MLP (F1 score of 0.992 for hello flooding, 0.988 for decrease rank attack, and 0.974 for version number attack). The techniques proposed so far for anomaly detection in RPL are supervised machine learning techniques.

3. System model and research gaps

Consider an IoMT network in which all the nodes in the network are required to transmit information to the coordinator or root node in a convergecast manner. To facilitate this, every node is required to maintain a route to the root node in a proactive manner such that when data has to be transmitted, the route is already available. The nodes in the network can be static or mobile, which may act as both generator and forwarder of data. All the nodes support Internet Protocol version 6 (IPv6) communication and form a low speed (in the range of 0–10 m/s) and low-power wireless personal area network. The nodes may not be in the direct range of the root node and hence the data will be transmitted by the nodes to the root node in a single or multi-hop manner till it reaches the root node. The root node may transmit the information over a local area or a wide area network. Typical use cases include a group of robots performing a surveillance task to identify mines in an area or a group of drones performing a search and rescue mission or a group of robots in a warehouse handling the management of objects in a warehouse, asset management in the healthcare applications, etc.

The widely used RPL routing protocol is primarily designed for static IoT networks. Based on the literature review presented in Section 2, the following gaps have been identified in enhancing RPL to handle mobility in the network: **(1) Reducing network disconnection time due to mobility:** In RPL, to reduce the overhead caused by the transmission of the repeated DIO messages, the duration of the trickle timer is increased exponentially by the nodes if no topological changes in the network are observed by it. However, due to this, if a node moves away from the current position or leaves the DODAG, it will take a longer duration for the nodes to observe the non-receipt of DIO from the parent, and this will delay the route maintenance process, leading to route breakages and packet loss. **(2) Reducing communication overhead and global DoDAG updation due to mobility:** The trickle period of the node in RPL is reset to the trickle timer's minimum duration on the detection of any inconsistency in the network, which results in a large communication overhead due to frequent transmission of DIO messages. When mobile nodes are present in the network, and connectivity with the parent node is lost, the RPL's trickle timer resets. This reset triggers the global updation of the DODAG and increases the control overhead within the network. For networks with mobile nodes, there is a need for a mechanism that leads to only local updates of the parent list, confined to nodes affected by mobility, rather than updating the entire network. **(3) Swift transition to a new Parent:** A node has the option to send DIS and trigger the immediate transmission of DIO from surrounding nodes that are already a part of the DODAG. The RPL protocol does not clearly define the time of sending DIS. Identification of mobility is a crucial aspect for deciding the time of transmission of the DIS and for minimizing the communication overhead in the network. Proactive methods are required to identify that the node and the current parent node are suffering from weak connectivity due to mobility and ensure a smooth transition from the current parent to another parent before disconnection from the current parent. The available works in the literature require the transmission of additional control messages for discovering mobility. Mechanisms are required in RPL to detect mobility without incurring additional communication overhead. **(4) Mobile nodes as forwarder nodes:** Most of the works in the literature on improving the RPL to handle mobility have assumed that only the leaf nodes (which only generates data but does not forward the data) are mobile in the network. However, more work is required to accommodate the mobility of nodes which are generators and forwarders of data. **(5) Resilience towards routing attacks:** Supervised machine learning techniques, that requires labeled traffic under normal and attack scenarios for training, are not suitable for anomaly identification and isolation especially when nodes are mobile. Unsupervised techniques for the identification of routing attacks are required to be developed and mechanisms to isolate the malicious nodes from the network while ensuring that the rest of the network continues to function without disruption are required.

In this work, we propose a modified version of RPL, named RPL*, designed to effectively handle node mobility while enhancing resilience against routing attacks. RPL* presents resilience towards routing attacks as the protocol has the ability to detect attacks and recover from attacks through the isolation of malicious nodes. In RPL*, any deviation from the normal behaviors of the network are identified as anomalies using an unsupervised Explainable Artificial Intelligence (XAI) strategy. The routing attacks will be identified as anomalies, and the affected nodes will be isolated, following which the rest of the IoT network, will continue to operate normally.

4. Proposed architecture - RPL*

This section elaborates the proposed routing protocol RPL* which is suitable for handling mobility in RPL networks and is resilient against routing attacks. RPL* is an enhanced version of RPL that includes the following (1) modifications to support mobility of nodes in the network (2) modifications to detect anomalies, present if any in the network. In RPL*, the proposed modification to support mobility will be executed on all the nodes and the modification to support anomaly isolation will be executed at the root node. The modifications incorporated to handle mobility and anomaly are elaborated in Sections 4.1 and 4.2 respectively. The major modifications in RPL* when compared to RPL are highlighted in Fig. 1 and is explained in the following sections.

4.1. Mobility management in RPL

Unlike previous works that assume mobile nodes do not participate in routing and forwarding, our approach considers that mobile nodes in the network can perform routing and forwarding roles in addition to data generation and collection. When nodes involved in routing and forwarding are mobile, the likelihood of frequent network connections and disconnections increases significantly. As mentioned in Section 2, the works in literature typically detects node mobility through the transmission and reception of additional control messages. Detecting node mobility without incurring these additional messages presents a challenge. Our modifications to support mobility in RPL* are achieved through three components: priority setting, parent selection, and connectivity management.

As shown in Fig. 1, we have introduced a periodic timer, termed the Mobility Timer (M-Timer), to handle mobility in RPL*. Upon the expiry of this timer, nodes are assigned a priority as detailed in this subsection. This allows us to detect mobility without incurring additional control messages. Additionally, we have implemented a connectivity management mechanism which will check the connectivity with the parent node on the identification of mobility in the network and trigger parent discovery mechanism to ensure that the node can be connected to another parent at the earliest if required due to mobility. The processes of priority setting, parent selection, and connectivity management in RPL* are elaborated below.

Priority setting of a node based on mobility: In RPL, the route updates are triggered on the receipt of DIO message by a node. The transmission of DIO occurs on the expiry of the trickle timer (the interval of timer is increased if the network is stable). To account for the mobility we have introduced an additional timer called Mobility-timer (M-timer) which is an auto-reload timer based on a configurable interval. The interval is set as 1 s in our work although the duration can be increased or decreased based on the velocity of the nodes in the network. The interval of M-timer has to be smaller than the minimum duration of the trickle timer in

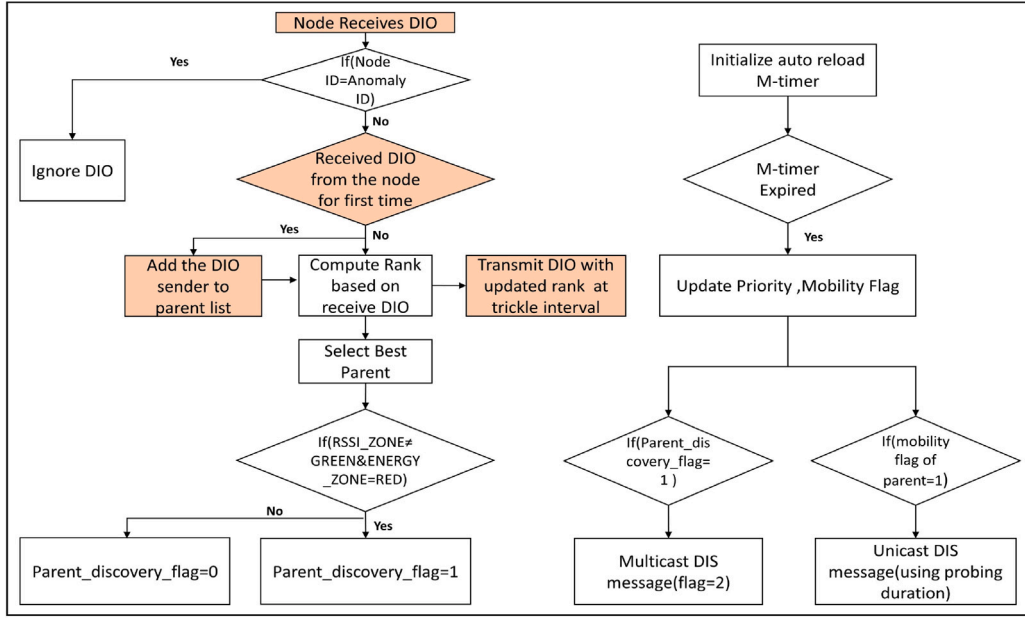


Fig. 1. Overview of modifications introduced in RPL (aspects in original RPL are in shaded boxes).

RPL. Upon the expiry of the M-timer, the priority setting and new parent discovery procedures are executed. The priority setting and new parent setting procedures are elaborated in Algorithm 1.

The priority setting procedure assigns a node's priority based on detected mobility. Based on the intuition that mobility in the network, causes the nodes to change their parent nodes more frequently than the static nodes, we have designed a mechanism to set a mobility flag in a node which changes its parents frequently (either due to own mobility or due to the mobility of its parent node). As indicated in Fig. 2, the mobility flag field is introduced into the DIO message so that the neighbor nodes can consider the same while performing the parent selection. The nodes with the mobility flag set is assigned a lower priority while being considered as a parent node.

As described in Algorithm 1, the priority setting procedure maintains the elapsed time since the last parent switch in a variable called as $t_{elapsed}$. The value of $t_{elapsed}$ is incremented on the expiry of M-timer, and is reset to 'zero' on a parent switch. A time window t_{window} as mentioned in the Algorithm 1 is calculated as the weighted moving average of the parent analysis time t_{pa} and $t_{elapsed}$. The initial parent analysis time t_{pa} window is set as twice the initial interval of the trickle timer in RPL (i.e. $t_{pa} = 2^{DIO_{min}+1}$ ms). The t_{window} is updated on a parent switch or when the t_{window} reaches '0'. This value of t_{window} is assigned to t_{mc} , the mobility check value. The t_{window} is decremented on every expiry of M-timer till it reaches '0'. When t_{window} reaches '0' the t_{window} is recalculated based on t_{pa} and $t_{elapsed}$, even if no parent switch observed, ensuring that the t_{window} is increased if the parent switch is not observed for a longer duration. As mentioned in Algorithm 1, if t_{mc} is less than the threshold to detect mobility, i.e. th_m , the mobility flag is set and the priority is set to 'zero'. The value of the threshold th_m is based on the velocity and transmission range of nodes. The value of th_m used in our performance analysis is explained in Section 5. The mobility flag and priority of the node are updated on every expiry of 'M-timer' based on the t_{window} . The calculation of t_{window} as the weighted moving average of the t_{pa} and $t_{elapsed}$ helps prevent incorrect setting of the mobility flag due to factors other than mobility in the network, such as a parent change because the current parent is running out of battery. The parent selection process summarized in Algorithm 2 is elaborated later in this section.

The proposed technique is also scalable, i.e. robust to the addition of nodes in the network. If a static node is introduced in the network and does not observe parent changes frequently, its mobility flag will be set to '0' and will also be eligible for becoming the best parent if required in the network. Conversely, if the newly introduced node is assigned the mobility flag of '1', it will only become the best parent if no neighboring nodes with a mobility flag set to 'zero' are available. The notable feature of this novel procedure is that the mobility detection can be performed without the transmission of control messages.

Parent Selection: In RPL, the nodes are assigned a rank based on the objective functions, Objective Function Zero (OF0) and Minimum Rank with Hysteresis Objective Function (MRHOF). MRHOF uses the Expected number of Transmissions (ETX) as its metric whereas the minimum hop count is used as the metric in OF0. ETX is the minimum number of transmissions required to deliver a packet successfully from a node to the root node in a DODAG. ETX, may fluctuate significantly under mobility conditions. Hence, to stabilize ranking, we have utilized only hop count as the objective function to determine the rank of the node. We then utilize the residual energy of the node to be considered as the preferred parent and received RSSI measured at the node to select

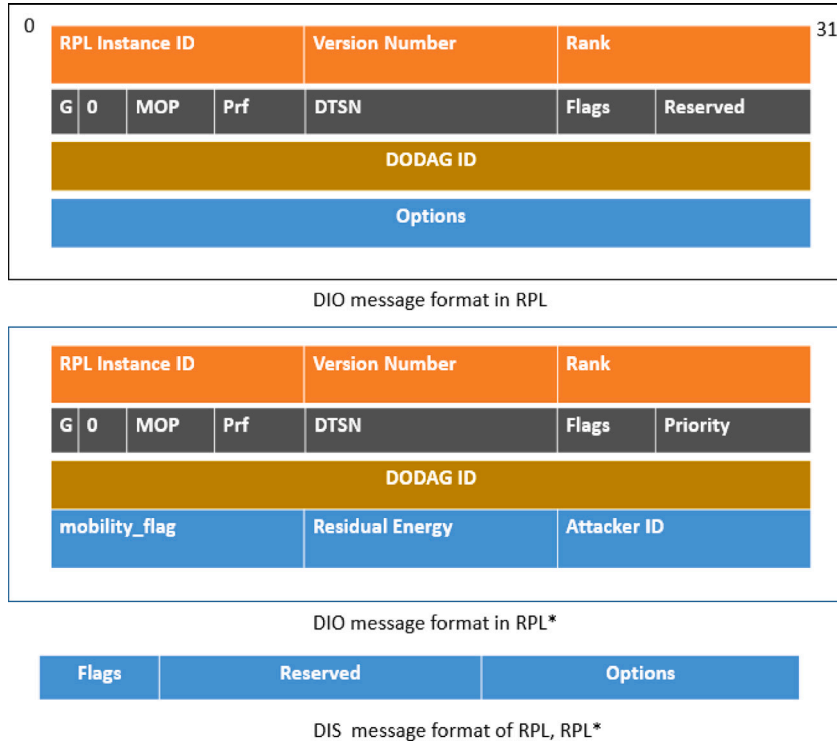


Fig. 2. DIO and DIS message formats in RPL and RPL*.

the best parent among the nodes with the same rank in the parent list. The RSSI of the received DIO message from a node can be calculated as

$$RSSI(dBm) = \eta \times 10 \times \log(d) + P_0 \quad (1)$$

where ' η ' is the path loss factor, ' d ' is the distance between the two nodes and ' P_0 ' is the measured power in dBm between two nodes, if they are kept at a distance of 1 m. Energy consumption is a critical aspect in low power and lossy networks. The major source of energy consumption in a mote is the communication cost comprising of sending and receiving the packets in the network. The energy consumption for the sky mote which is used in our work was calculated as:

$$E_{con} = \frac{[tx \times 19.5 + rx \times 21.8 + CPU \times 1.8 + LPM \times 0.0545]mA \times 3V}{32768} \quad (2)$$

where 'tx', 'rx', 'CPU' and 'LPM' are the total number of ticks expended in the transmit mode, receive mode, active mode and in the low power mode of the skymote after the power on. The skymote was operated using a 3 V power supply and at a clock frequency of 32768 Hz. The residual energy (E_r) is obtained by subtracting the energy consumption (E_{con}) from the initial energy of the node E_t .

$$E_r = E_t - E_{con}$$

Powertrace plugin in Cooja was used for the measurement of power consumption. As the instantaneous RSSI and residual energy values will fluctuate, instead of using the absolute values of RSSI and residual energy, we have mapped the RSSI and residual energy values of nodes into three categories/zones as shown in Table 1 and Table 2. We then utilize the residual energy transmitted by the node to be considered as the preferred parent and received RSSI measured at the node to select the best parent among the nodes with the same rank. The residual energy is transmitted in the DIO field as indicated in Fig. 2. Algorithm 2 provides the pseudo code for the best parent selection from the nodes in the parent list. All possible pairs of nodes in the parent list are evaluated for the best parent selection.

Connectivity monitoring Mechanism: The connectivity monitoring mechanism primarily consists of the novel parent discovery process which is triggered on the expiry of a M-timer. This ensures that at an interval of M-timer, the connectivity with the parent node is checked through the transmission of DIS messages by a node if the '*mobility_flag*' of the node is set or the '*mobility_flag*' of its parent is set. Notably, RPL lacks the implementation of a proactive connectivity mechanism. The proposed novel parent discovery mechanism procedure is explained in Algorithm1. The DIS message format is as shown in Fig. 2. When the '*mobility_flag*' of a node or its parent is set, the node transmits (unicasts) the DIS message to its current parent with a 'Flags' field set to '1'. Upon receiving

Algorithm 1 Procedures executed on the expiry of the ‘M-Timer’

```

Initial values:  $mobility\_flag=1$ ,  $t_{pa} = 2^{DIO_{min}+1}ms = 8s$ ,  $t_{elapsed} = 0$ 
procedure PRIORITY_SETTING
   $t_{elapsed} += 1$ 
  if (PARENT_SWITCH=TRUE) then
     $t_{window} = \alpha \times t_{pa} + (1 - \alpha) \times t_{elapsed}$ 
     $t_{pa} = t_{mc} = t_{window}$ 
     $t_{elapsed} = 0$ 
  else if  $t_{window} > 0$  then
     $t_{window} -= 1$ 
  end if
  if  $t_{window} == 0$  then
     $t_{window} = \alpha \times t_{pa} + (1 - \alpha) \times t_{elapsed}$ 
     $t_{mc} = t_{window}$ 
  end if
  if  $t_{mc} > th_m$  then
     $mobility\_flag = 0$ ;  $Priority = 1$ 
  else
     $mobility\_flag = 1$ ;  $Priority = 0$ 
  end if
end procedure
procedure NEW PARENT DISCOVERY
  if ( $mobility\_flag = 1$ ) then
    for  $n \leq N$ 
      Transmit unicast DIS message to preferred parent with ‘Flags’ field set to ‘1’
      Wait for DIS Interval, Increment N
    end for
  else if ( $mobility\_flag$  of parent == 1) then
    for  $n \leq N$ 
      Transmit unicast DIS message to preferred parent with ‘Flags’ field set to ‘1’
      Wait for DIS Interval, Increment N
    end for
  end if
  if ( $parent\_discovery\_flag = 1$ ) then
    Transmit broadcast DIS message with ‘Flags’ field set to ‘2’
    Wait for DIS Interval, Increment N
  end if
   $N=0$ ;
end procedure

```

Table 1
RSSI zone selection in RPL*

Condition	RSSI-ZONE	Value
$RSSI \leq Rssi_{ih1} $	GREEN	2
$ Rssi_{ih1} < RSSI < Rssi_{ih2} $	Gray	1
$RSSI \geq Rssi_{ih2} $	RED	0

Table 2
Energy zone selection in RPL*

Condition	ENERGY_ZONE	Value
$E_r \geq 0.5 E_i$	GREEN	2
$0.3 E_i \leq E_r < 0.5 E_i$	GRAY	1
$E_r < 0.3 E_i$	RED	0

this, the parent node transmits a DIO message, initiating a parent selection based on our parent selection scheme. If the two nodes remains within the ‘GREEN’ zone of the RSSI, the current parent will continue. However, if the two nodes are in the ‘GRAY’ or ‘RED’ zone of RSSI, a new parent selection will be attempted by the node. Prior works in the literature utilize reactive methods to

Algorithm 2 Parent selection technique

P1, P2 -nodes in the parent list, rssi- RSSI recorded from the node in the parent list under consideration, energy- residual energy left in the node being considered as the parent.

procedure BEST_PARENT(P1, P2)

 r1=calculate Rank of P1

 r2=calculate Rank of P2

 Call Zone Selection procedure for nodes P1 and P2

 Select the best parent based on Table 3

 For the selected best parent node, set the *parent_discovery_flag* = 1 if its RSSI_ZONE is GRAY or RED and if the ENERGY_ZONE is RED. Else, set *parent_discovery_flag* = 0.

end procedure

procedure ZONE_SELECTION(*p*, *rssi*, *energy*)

if $p \rightarrow rssi \leq |Rssi_{th1}|$ **then**

$p \rightarrow$ RSSI_ZONE = GREEN

else if $|Rssi_{th1}| < p \rightarrow rssi < |Rssi_{th2}|$ **then**

$p \rightarrow$ RSSI_ZONE = GRAY

else

$p \rightarrow$ RSSI_ZONE = RED

end if

if $p \rightarrow energy \geq 0.5 \times E_t$ **then**

$p \rightarrow$ ENERGY_ZONE = GREEN

else if $0.3 \times E_t < p \rightarrow energy < 0.5 \times E_t$ **then**

$p \rightarrow$ ENERGY_ZONE = GRAY

else

$p \rightarrow$ ENERGY_ZONE = RED

end if

end procedure

trigger DIO transmission from neighbor nodes when the nodes are disconnected from the parent. Such methods lead to a drop in the PDR due to the time required to update the route towards the root node via DODAG updation. In our novel parent discovery mechanism, the nodes (static or mobile node) suffering from weak connectivity with its current parent (e.g: nodes which lie in the 'RED' or 'GRAY' zone of RSSI or the 'RED' zone of the residual energy) transmit DIS message (with the 'Flags' field is set to '2') in a broadcast manner. The proposed technique is a proactive method such that the DIS message can be sent to the neighbor nodes before the connection with the parent is lost, thus triggering the joining of the child to another parent in the parent list. This proactive approach ensures DIS messages can potentially establish new parent-child links before disconnections occur, enhancing network resilience. Nodes other than the preferred parent respond with a DIO. After sending the DIO, tickle timer is not reset to ensure that the complete DODAG updation is not triggered and only the new parent is selected for the nodes as a result of the new parent discovery mechanism. The complete updation of DODAG will occur based on trickle timer as it occurs in RPL. The carefully designed parent discovery process through the transmission of the DIS and DIO by only relevant nodes reduces the communication overhead when compared to the RPL. The proactive mechanism also prevents the partitioning of the network, reduces the network disconnection time and improves the PDR.

In RPL*, DIS probing is conducted over a specified probing duration to assess connectivity between parent and child nodes, especially when either or both nodes are mobile. Within the probing duration, the connectivity is checked for a varying number of DIS messages transmitted at an equal interval (referred to as DIS interval, $DIS\ Interval = \frac{DIS\ probe\ duration}{N}$ within the probing duration as mentioned in Table 4. We have tested the network for various values of DIS interval and tested the impact on PDR, and control traffic overhead on RPL* to decide on the optimal number of DIS probing packets within the interval. The results are summarized in Section 5.

4.2. Anomaly detection using explainable AI (XAI)

Supervised machine learning techniques that require labeled traffic under normal operational conditions and the traffic affected by attack/anomalies for training the model are not appropriate for the anomaly identification and isolation in practical scenarios. To address this, we have designed an Explainable AI (XAI) based unsupervised technique for the detection of anomalous behaviors in the RPL networks. In the proposed RPL*, any deviation from the normal behaviors of the network are identified as anomalies using an unsupervised Explainable Artificial Intelligence (XAI) strategy. This method effectively identifies routing attacks as anomalies, enabling the isolation of affected nodes and facilitating network maintenance. Our technique leverages the Isolation Forest (IForest), an unsupervised machine learning model, complemented by a rule extraction procedure for anomaly detection within the XAI

Table 3
Case table for best parent selection in RPL*

Condition	Condition	Return
$r1 > r2$	–	P2
$r2 > r1$	–	P1
$r1 = r2$	P2 $\rightarrow$ Priority > P1 $\rightarrow$ Priority	P2
	P1 $\rightarrow$ Priority > P2 $\rightarrow$ Priority	P1
	P1 $\rightarrow$ RSSI_ZONE > P2 $\rightarrow$ RSSI_ZONE and (P1 $\rightarrow$ ENERGY_ZONE > 0 P2 $\rightarrow$ ENERGY_ZONE > 0)	P1
	P2 $\rightarrow$ RSSI_ZONE > P1 $\rightarrow$ RSSI_ZONE and (P1 $\rightarrow$ ENERGY_ZONE > 0 P2 $\rightarrow$ ENERGY_ZONE > 0)	P2
	P2 $\rightarrow$ RSSI_ZONE = P1 $\rightarrow$ RSSI_ZONE and P1 $\rightarrow$ ENERGY_ZONE > P2 $\rightarrow$ ENERGY_ZONE	P1
	Default:	P2

Table 4
Scenarios used for testing DIS probing.

Scenario	DIS probe duration (s)	N
a	16	1
b	16	2
c	16	3
d	32	1
e	32	2
f	32	3
g	64	1
h	64	2
i	64	3

framework. Unlike supervised approaches, IForest can be trained using unlabeled data encompassing both normal and anomalous traffic (such as traffic affected by attacks), ensuring robust anomaly detection without the need for labeled datasets.

In this work, we evaluated the effectiveness of RPL* against two prominent types of routing attacks, i.e the black hole and sink hole attacks. In blackhole attack, the malicious nodes drop the received data packets (whose destination is another node in the network) instead of forwarding the packets to the next node. In sinkhole attack, the attacker attracts more traffic towards it and subsequently drops the packets instead of forwarding it. In this work, we have introduced the sinkhole attack through rank decrease attack. In rank decrease attack, the malicious node announces its rank to be closer to the sink node than actually it is, attracts more nodes towards it, and then drops the data packets that are sent to it. IForest is an aggregation of distinct random trees was employed to distinguish anomalous traffic from normal traffic [32]. Each tree isolates the traffic by branching from the root node by selecting a random feature from the list of features in the input data. To branch from one node, IForest randomly chooses an attribute 'I', from the 'm' data attributes and chooses a threshold value 'v', between the minimum and the maximum value of 'I'. If the value of the feature in the input is less than 'v', the data follows the left branch else it follows the right branch. IForest iterates the process of tree updation until the complete input data set is isolated or until a maximal tree depth is reached through recursive partitioning. The path length of a tree is equal to the number of splits required to isolate a sample by a tree from the root node. Generally the path length will be smaller for anomalous samples.

Features Extraction for Anomaly detection: Datasets of RPL networks under attack in which nodes are also mobile are not readily available for training the model. To address this gap, we utilized the Cooja simulator to generate a dataset for evaluating RPL and RPL*. The radio traffic generated in the network is captured using Wireshark packet sniffer tool and a PCAP file is generated. The PCAP file is converted to CSV file using the Wireshark tool and fed into the pre-processing module. In the pre-processing phase, the node's source and destination IP addresses are replaced with the corresponding node's name of identifier (example: fe80::212:7402:2:202 is replaced with Node 2). After the pre-processing, the updated CSV files are then subjected to feature extraction using a Python script. For the detection of the specific attacks under consideration, features were extracted over a sliding time window of 30 s per node, spanning a total observation period of 3600 s.

- **Number of Parent Changes (NPC):** The number of parent changes by a particular node in the time window.
- **Number of child nodes (NC):** The number of child nodes connected to a node in the time window.
- **Number of rank change (NRC):** A node's rank is a specific field in the DIO packet indicates the node's position in a DODAG relative to its root node. Rank value will be reduced by the malicious node in order to declare a lower rank than neighbors, causing the DODAG to be modified. This principle is used in the rank decrease and sinkhole attacks. Hence, the number of rank changes in DIO messages in the window is also utilized as a feature for anomaly detection.
- **Number of UDP messages transmitted (UDP_transmitted):** The number of UDP messages generated by a particular node in the time window.
- **Number of UDP messages received (UDP_received):** The number of UDP messages received by a particular node in the time window.
- **Number of UDP messages forwarded (UDP_forwarded):** The total number of packets forwarded by the node in the time window.

Algorithm 3 Rule Extraction Algorithm

Require: Dataset D , Number of Estimators $-n$, Random State $-r$, Max Samples $-m$, Features Name $-F$, Minimum Recall $-R=0.7$, Minimum Precision $-P=0.8$

- 1: **procedure** RULE EXTRACTION
- 2: Split the data into training set $(X_{\text{train}}, Y_{\text{train}})$ and testing set $(X_{\text{test}}, Y_{\text{test}})$
- 3: Train the IForest model : $model \leftarrow \text{IsolationForest}(n, r, m)$; $model.fit(X_{\text{train}})$
- 4: Extract the rules from each estimator. Select the rules that provide precision and recall above a set threshold ($R=0.7$, $P=0.8$) for anomalous traffic in the training dataset and store rules in the rules dictionary: $rules, rule_dict \leftarrow rule_extract(model, F, X_{\text{train}}, Y_{\text{train}}, R, P)$
- 5: Apply each rule to the test data and calculate the prediction based on the rules. $y_{\text{pred}} \leftarrow rule_votes(X_{\text{test}}, rules)$
- 6: If any of the rules detect anomaly for the test data, the data is considered as an anomaly.
- 7: **end procedure**

Table 5
Different scenarios for evaluating the performance of RPL*

Scenario	Priority setting	Parent selection	Connectivity management
1 (RPL)	X	X	X
2	X	✓	X
3	✓	✓	X
4 (RPL*)	✓	✓	✓

- **Packet forward percentage (Packet_forwarding):** Percentage of packets forwarded by the node in the time window.

Each feature in the dataset is then normalized using the min–max scalar technique. The training data was collected for the network deployment as presented in Section 5.

In IForest, a scoring method is typically employed to classify data as anomalous or normal [32]. This method utilizes path length, assuming that anomalous samples will have shorter path lengths. However, various studies have identified the limitations of the scoring method, which lead to the misclassification of anomalous data as normal data and vice-versa [32,33]. The scoring method exhibits less sensitivity to outliers that are close to normal data points, resulting in misclassification of anomalous and normal data [32]. Moreover, the scoring method fails if there are a large number of similar anomalies in which a longer path length may be required to separate the anomalous data during training [33]. Hence, to overcome the limitations of the scoring method and to ensure the explainability of the model, we have applied rule extraction technique on the trained IForest for the identification of the abnormal traffic as mentioned in Algorithm 3. The input features from the dataset are fed to the trained model and the rules are extracted from each tree. During inference, these rules are applied to aggregated incoming traffic captured over a time window at each node, enabling classification of the traffic as normal or anomalous.

RPL* assigns less priority to the mobile node for being selected as a parent. If only the mobile node(s) are available for the selection as a parent node and the node moves away from the transmission range for some duration, it may drop the packets. The nodes may drop traffic due to the mobility and hence the nodes which are reported as anomalous are not immediately removed as a parent from DODAG. Instead, these abnormal nodes are added to an anomaly list maintained by the root node for further monitoring. RPL* also implements connectivity monitoring mechanism, to swiftly recover from temporary disconnections. Nodes listed as anomalous by the XAI technique for three consecutive monitoring windows are declared anomalous. Conversely, nodes not flagged as anomalous for three consecutive windows are removed from the anomaly list. Upon detection of an anomalous node, the root node sends a multicast DIO message to the nodes in the network by adding the node ID of the attacker node in the message as indicated in Fig. 2. The node ID of the attacker node will be transmitted further by all receiving nodes in their DIO message. Upon receipt of this message, any node with the anomalous node in its parent list removes it from the list. If the malicious node is the current parent, the nodes will initiate the new parent discovery by sending a DIS message.

5. Implementation and performance evaluation

We have implemented RPL and RPL* on Cooja Simulator on Contiki OS version 3.0. We have utilized NullRDC, the always-on radio model at the link layer for the nodes. The mobility management in RPL* consists of three major modules — priority setting, parent selection and connectivity management. We have examined the performance of the RPL* algorithm for different scenarios as mentioned in Table 5. Scenario 1 in Table 5 represents the conventional RPL technique. Scenario 2 represents RPL with modification in the parent selection as introduced in RPL*. Scenario 3 represents RPL with the modification in the parent selection and priority setting mechanism as introduced in RPL*. Scenario 4 represents the proposed RPL* technique with the modification in the parent selection, priority setting mechanism and connectivity management modules. We have tested RPL* for various mobility models, network topologies, and varying number of mobile nodes as presented in the following subsections.

Table 6
Mobility model parameters.

Mobility Model	Simulation parameter
RWP	d - simulation time, n- number of nodes, l- starting delay= 0, (x,y) - maximum x,y coordinate, p- pause time, h- maximum speed , l- lowest speed
MM	d - simulation time, n- number of nodes, l- starting delay= 0, 'e' is the minimum speed,'m' is mean speed,'o' is maximum pause time,'u' number of blocks along the x-axis and 'v' number of blocks along y axis.
NCM	d - simulation time, n-number of nodes, l starting delay= 0, (x,y) - (highest x,y coordinate), (x,y) - maximum x,y coordinate ,p pause time, -h maximum speed = 3, -l lowest speed =3, a-the average number of nodes per group=2, r -the maximum distance from the group center=10

5.1. Mobility models

We have tested RPL* on three popular mobility models, i.e. Random Way Point (RWP) model, Manhattan Mobility (MM) model and Nomadic Community Mobility (NCM) model [34].

RWP mobility model proposed by Johnson and Maltz is widely utilized for the evaluation of Mobile Ad-hoc network (MANET) routing protocols. In RWP, each mobile node randomly selects a destination in the simulation area and starts moving towards the destination with random velocity selected from the window $[0 - v_{max}]$, where ' v_{max} ' is the maximum speed. The direction and velocity selected by a node are independent of other nodes. When the node reaches the destination, it stops for a duration of time (i.e. the pause time) and chooses another random destination in the deployment area and moves towards it with the newly selected random velocity. This process is repeated till the end of simulation.

In Manhattan Mobility (MM) model, the mobile nodes are fed with a predefined map of an area described using horizontal and vertical lines or grids [34]. The mobile node can choose to go straight or turn right or left at an intersection of a horizontal and a vertical line, with a certain probability. This mobility model is well suited for applications such as self-driven vehicles.

In Nomadic Community Mobility (NCM) model, a fixed number of nodes move in a group and the group members move to the destination using the random walk mobility model [34]. This model can be employed in applications in which a group of nodes perform similar operations in a collaborative manner in the area of deployment. e.g. first responder robots in a mine, robots deployed in surveillance, etc.

We have tested RPL* using the above mentioned mobility models such that the network can be used for the random mobility, structured mobility as well as group mobility thus catering to a wide range of applications. The mobility traces for these models are generated using the BonnMotion tool [34]. The configurable parameters for each mobility model and their values used for the generation of the mobility trace using Bonn Motion tool for RWP, MM and NCM models are mentioned in Table 6.

5.2. Performance metrics

We have utilized the metrics, Packet Delivery Ratio (PDR), end-to-end latency, network disconnection time and control packet overhead for the performance evaluation of RPL* .

Packet Delivery Ratio (PDR): is the ratio of the number of packets successfully received (N_r) from a node, at the root node to the total number of packet transmitted (N_t) by the node in the network.

End-to-End Latency: is measured as a difference between the time at which the packet was sent from the source ($S_t(n)$) and the time at which it was received at the root node ($R_t(n)$). The average latency (T_l) for ' N ' packets is calculated as

$$T_l = \frac{\sum_{n=1}^N [R_t(n) - S_t(n)]}{N} \quad (3)$$

where ' $R_t(n)$ ' is receive time and ' $S_t(n)$ ' is the sent time of the ' n^{th} ' packet.

Control Traffic Overhead (CTO): is calculated as the total number of DIS, DIO, DAO, DAO-ACK messages transmitted or received by all the nodes during the operational time of the network.

$$CTO = \sum_{n=1}^k [DIS(n), DIO(n), DAO(n), DAO_ACK(n)] \quad (4)$$

5.3. Performance analysis

A network comprising of sky motes was deployed for testing the RPL and RPL* protocols. The value of path loss factor ' η ' in equation (1) was determined as $10^{3.5}$ by placing the nodes at known distances. The nodes were configured in Cooja with a transmission power of -31 dB m and transmission range of 40 m. The $Rssi_{ih1}$ and $Rssi_{ih2}$ in Table 1 were configured considering the coverage reach of the node based on the transmission power. In our work, we have set the $Rssi_{ih1}$ as -82 dB m which is

Table 7
Simulation parameters.

Index	Simulation parameter	Values
1	Simulation time	360 s
2	Mobility	2-10 m/s
3	Data rate	250 kbps
4	Area	100 m × 100 m
5	No. of nodes	13
6	Type of Mote	Sky Mote
7	Channel frequency	2.4 GHz
8	Physical and Mac Layer	IEEE 802.15.4
9	Payload Size	30 bytes
10	Data periodicity	15 s
11	Mobile nodes	20%–50%
12	Topology	Random, Grid, Elliptical
13	Radio medium	Unit Disk Graph medium
14	Mobility Model	RWP, MM, NCM
15	Transmission Range	40 m
16	Interference Range	60 m

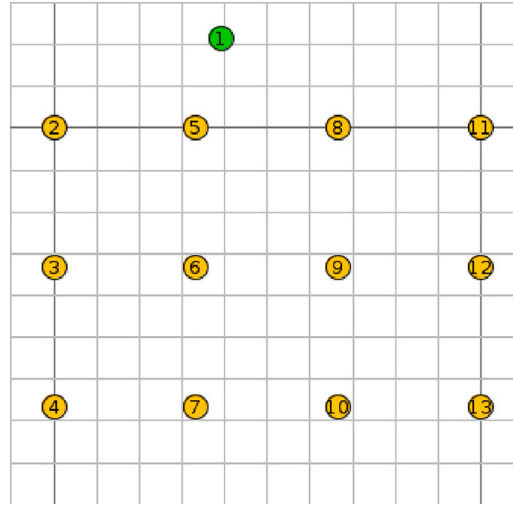


Fig. 3. Grid topology.

equivalent of 30.62 m distance between two nodes and $Rssi_{th2}$ as -85 dB m which corresponds to 35.75 m between the nodes. Also, the threshold ' th_m ' in Algorithm 1 is configured as 60 s for the simulation parameters mentioned in Table 7. All nodes were configured to send periodic traffic at a periodicity of 15 s to the root node. The various scenarios used for testing RPL* are presented below.

A network of 13 sky motes (in addition to the root node with ID '1') were deployed in the grid topology as shown in Fig. 3. The nodes were tested for all 4 scenarios as mentioned in Table 5. Scenario 1 in Table 5 represents the conventional RPL technique. Scenario 2 represents RPL with the proposed modification in parent selection as introduced in RPL*. Scenario 3 represents RPL with the proposed modification in the parent selection and priority setting mechanism in RPL*. Scenario 4 represents the RPL*, i.e RPL technique with the proposed modification in the parent selection, priority setting and connectivity management. The effect of each of the mechanisms introduced in the RPL* are elaborated in this section. The nodes are assumed to be moving in the formation as shown in Fig. 3. Such formations can be used by drones flying at the same altitude for search and rescue missions, robots searching for mines in an area, etc. RPL and RPL* protocols were tested with two nodes (with ID 4 and 10) with a Manhattan grid mobility model at a relative speed of 3 m/s from the other nodes and eleven nodes in static condition (i.e at a relative speed of 0 m/s with respect to each other, all moving in the same direction). Assuming ideal link conditions, the scenario is same as keeping the static nodes (at speed 0 m/s) and node 4 and 10 moving with a speed of 3 m/s. In Section 4, for further analysis, we have kept the static nodes (at speed = 0 m/s) and only mobile nodes move in the area of deployment. The simulation parameters are mentioned in Table 7. The results are presented in Figs. 6–9. As shown in Fig. 6, the average PDR of the RPL (Scenario 1) and RPL* (Scenario 4) for the static nodes in the network is 100%.

In RPL, nodes change their parent based on the minimum hop to the root node. However, in RPL*, the nodes change their parent if the new node under consideration as parent node falls within the higher RSSI zone or higher residual energy zone as mentioned in Table 3. The drop in PDR observed for static nodes in Scenario 2 in Fig. 6 is due to the variation in the RSSI between the nodes

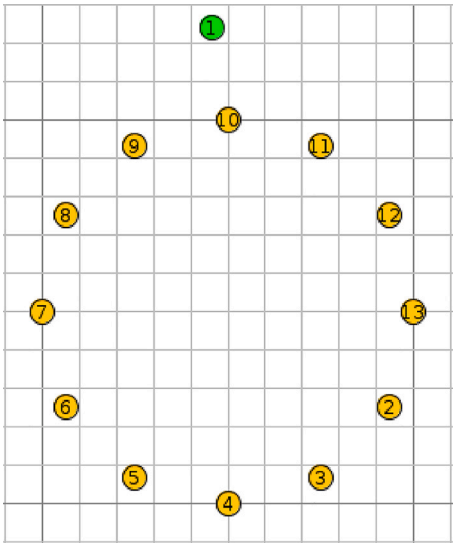


Fig. 4. Elliptical topology.

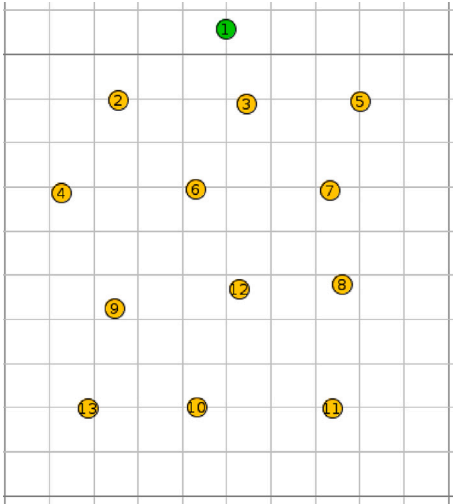


Fig. 5. Random topology.

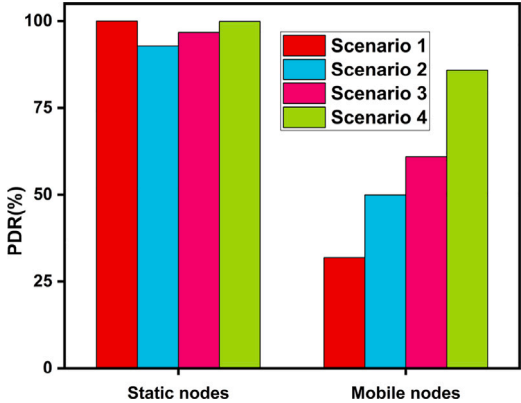


Fig. 6. Packet delivery ratio.

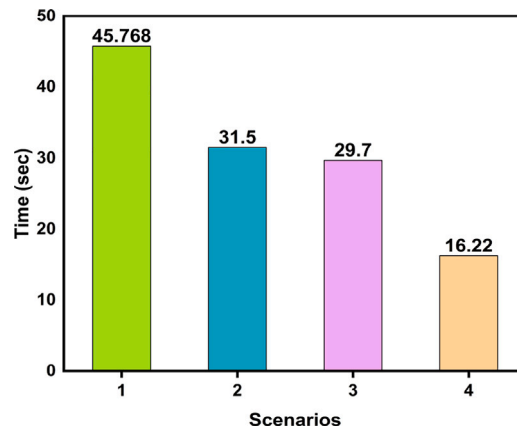


Fig. 7. Time to join the disconnected nodes to the network.

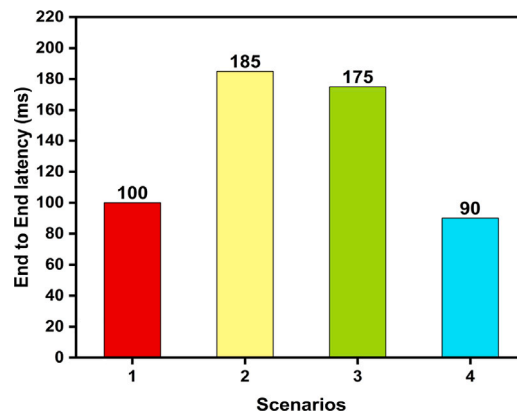


Fig. 8. End-to-end latency of mobile nodes.

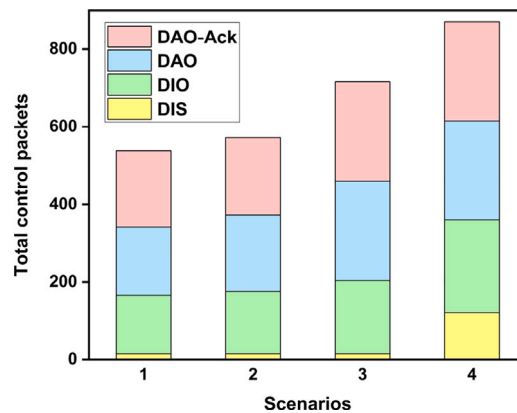


Fig. 9. Routing overhead.

because node 4 and 10 are mobile. The proposed parent selection method is required to handle mobility and it can be seen from Fig. 6 that the PDR of mobile nodes is increased when the parent selection scheme is introduced. In Scenario 3, it can be observed that due to the priority setting process, the parent discovery process is started as soon as a parent change is observed or when the parent node and the child node move out of the 'GREEN' zone of RSSI or the parent node is in the 'RED' zone of energy leading to an improvement in PDR for both static and mobile nodes. Hence, it is important to implement the priority setting based on mobility along with the parent selection process to achieve an overall improvement in PDR. This also leads to a reduced network

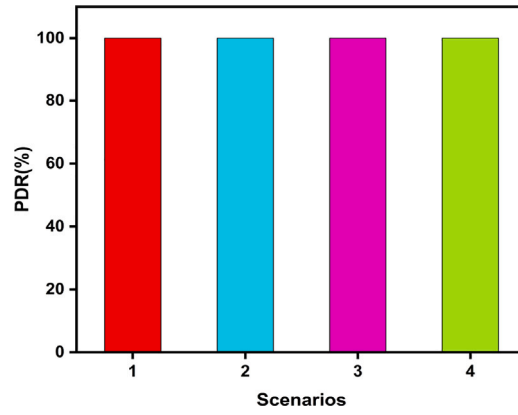


Fig. 10. PDR for the network in static condition.

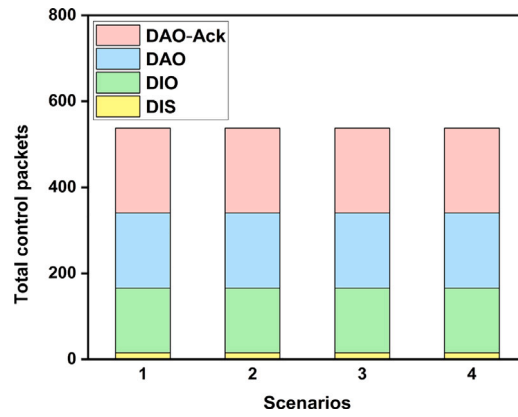


Fig. 11. Total control messages for the network in static condition.

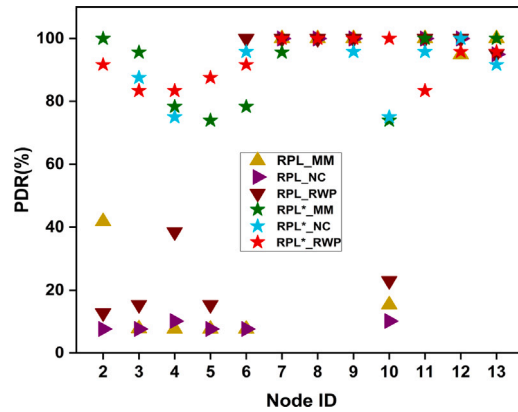


Fig. 12. PDR for elliptical topology.

disconnection time. From Fig. 6, it can be observed that the average PDR for mobile nodes i.e node 4 and 10 increases with the introduction of each of the introduced features such as priority setting, parent selection and connectivity management technique.

In RPL, the average PDR of mobile nodes was 31.5% in RPL; whereas RPL* provides a PDR of approximately 95%. From Fig. 7, it can be observed that with the introduction of priority setting, parent selection and connectivity management mechanisms, the network disconnection time for the mobile nodes is reduced. From Fig. 8, it can be observed that if connectivity management and priority setting mechanisms are not utilized, the end-to-end latency in packet delivery will increase. RPL* checks that nodes are connected to the parent using DIS probing when the mobility is detected. Even though there is an increase in the number of control

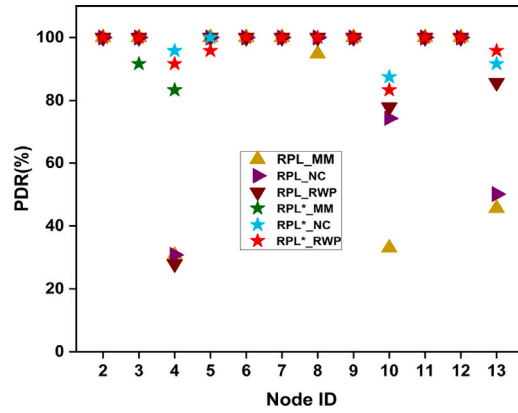


Fig. 13. PDR for grid topology.

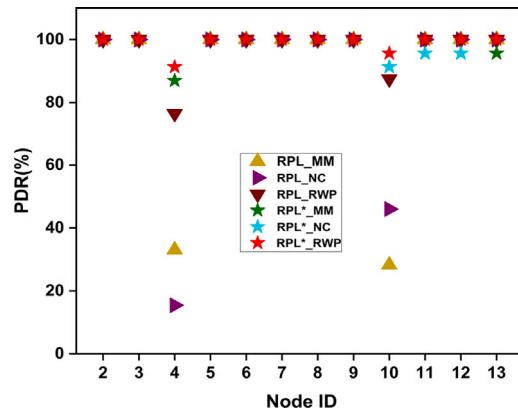


Fig. 14. PDR for random topology.

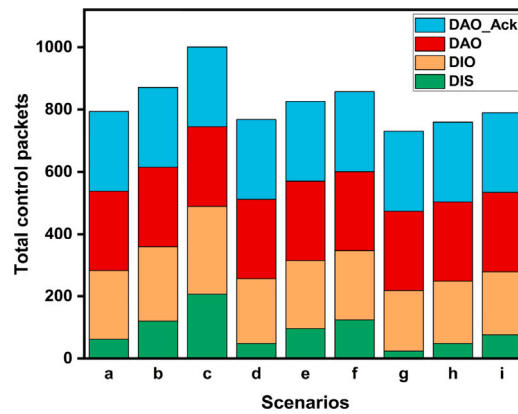


Fig. 15. Total control packets transmitted Vs DIS probing interval.

messages as shown in Fig. 9 for RPL* due to DIS probing, the other performance metrics such as average PDR, the end-to-end latency and the reconnection time are improved.

Undoubtedly it can be observed that RPL* improves the performance of the network under mobility. The RPL* protocol was also tested with all nodes under static condition. The results as shown in Figs. 10–11, indicate that the mechanisms introduced to support mobility does not lead to a drop in PDR or increase in control packet overhead.

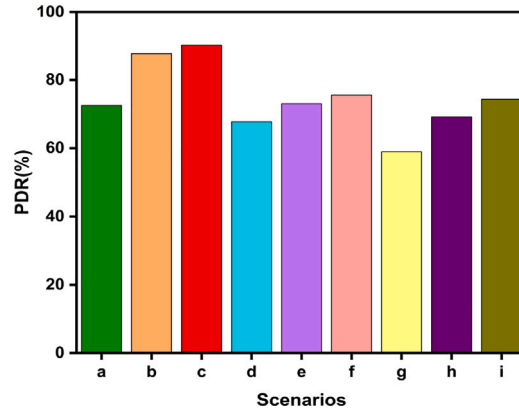


Fig. 16. PDR of mobile nodes Vs DIS probing interval.

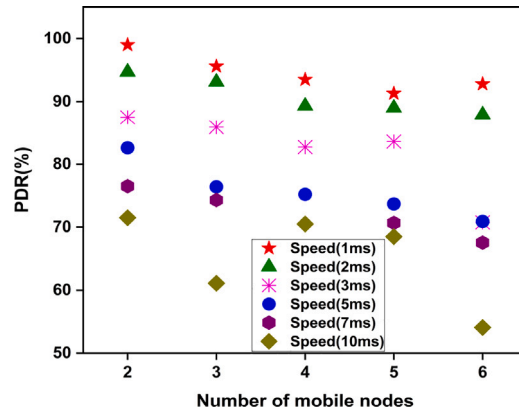


Fig. 17. PDR at various speeds (RPL*).

5.4. Impact of mobility model and topology

We have tested RPL* with different mobility models such as RWP, MM, and NC for different topologies such as grid, elliptical and random as shown in Figs. 3–5. Two nodes with node ID 4 and 10 were made mobile at a speed of 3 m/s. As per Figs. 12–14 the PDR of RPL protocol for mobile nodes 4 and 10 is significantly lower for all mobility models and topology when compared to RPL*. In RPL, the nodes change their parent if they receive DIO from a node with a better rank than previous parent or else they remain connected to that parent itself. If the parent or child node moves out of range of each other, the PDR will drop. In RPL, a new parent is identified only after losing connection with the existing parent. In Fig. 12, the PDR of static nodes 2,3,5 is observed to be lower for both RPL and RPL* because these nodes select the mobile node 10 as their parent, as it is closest and is connected to the root node. The parent selection in RPL* does not rely only on rank; nodes change their parents based on RSSI and energy. If the nodes are in the GRAY or RED zone of their RSSI, via DIS probing, the connectivity between the parent and child is ascertained and the parent discovery flag is set under weak connectivity. This ensures that the nodes can connect to a better parent (if available) before being disconnected from the current parent, which increases the PDR of the network.

5.5. Impact of DIS probing interval:

In RPL*, the DIS probing is done for a probing duration to check the connectivity between parent and child node when either or both of the nodes are mobile. Within the probing duration, the connectivity was checked for varying number of DIS messages transmitted at an equal interval (referred to as DIS interval, $\text{DIS Interval} = \frac{\text{DIS probe duration}}{N}$) as mentioned in Table 4. We have tested the network for various values of probing duration and evaluated the impact on PDR, and control traffic overhead (various scenarios are mentioned in Table 4) to decide on the optimal number of DIS probing packets within the interval. Smaller probing interval increases the overhead in terms of the control messages, however it can be observed from Figs. 15 and 16, that with the decrease in the DIS probing interval, PDR increases as expected. It may be noted that extremely shorter probing durations may not be effective as the increased control packet overhead may lead to collision and drop in the PDR in the network. As a balance between the reduction in disconnection time, improvement in PDR and the increase in control overhead due to DIS probing, we have selected the DIS interval of 16 s during which two DIS messages are transmitted in RPL*.

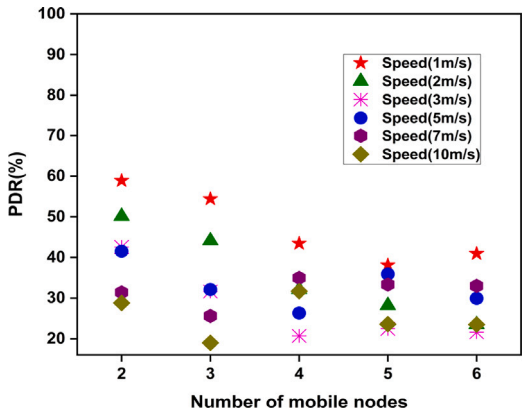


Fig. 18. PDR at various speeds (RPL)

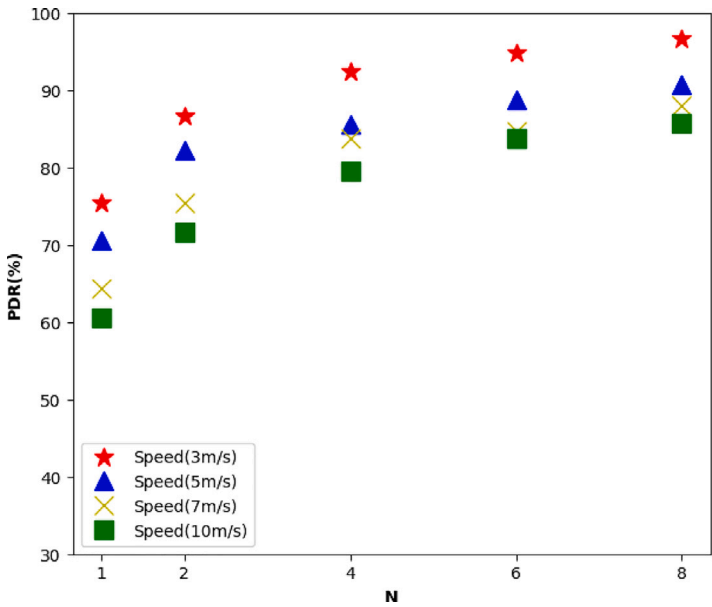


Fig. 19. PDR for various DIS intervals.

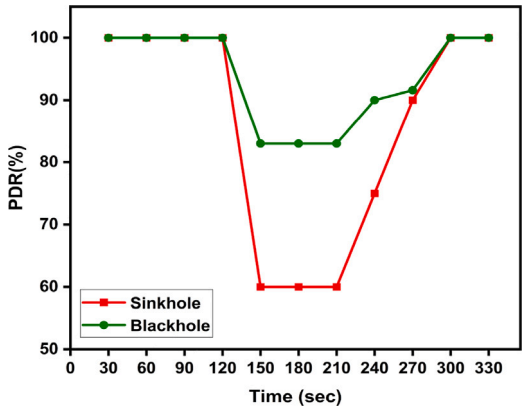


Fig. 20. RPL* PDR for static scenario.

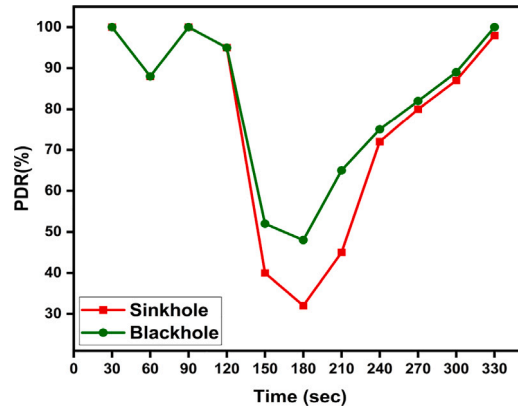


Fig. 21. RPL* PDR for mobile scenario.

Table 8

Node mobility and its value.

No of mobile nodes	Mobile node ID	Mobility Parameters
2	4,10	pause time = 20 s,
3	4,10,11	highest speed(h) = 10 m/s,
4	2,4,10,11	lowest speed(l) = 1 m/s,
5	2,4,6,10,11	Topology=Grid, mobility model = RWP
6	2,4,6,10,11,12	start time = 10 s

Table 9

Simulation parameters used for attack identification in RPL*.

Index	Simulation parameter	Values
1	Simulation time	3600 s
2	Time at which attack is introduced	120 s
3	node ID of mobile nodes	4,10
4	Transmission range	40 m
5	Interference range	100 m
6	Data periodicity	15 s
7	Payload length	30 bytes
8	Mobility introduced at	30 s

5.6. Impact of varying number of mobile nodes and varying speed

We have tested the proposed algorithm at various speeds and number of mobile nodes in grid topology using RWP mobility model as shown in Figs. 17 and 18. The ID of the mobile nodes and the simulation conditions are given in Table 8. The DIS probing duration was set to be 16 s and DIS interval is 8s. The performance of RPL* is better than RPL in all scenarios. It may be noted that the drop in PDR with increasing speed can be compensated with the reduction in DIS interval (i.e with an increase in 'N' mentioned in Table 4). We have tested the network for speeds from 3–10 m/s for varying values of 'N'. From Fig. 19, it can be observed that the increase in 'N' will lead to an increase in PDR. However, choosing a very high value of 'N' might lead to increase in control packets and drop in PDR.

5.7. Impact of routing attacks in the network

The proposed XAI based anomaly detection mechanism in RPL* was tested to detect anomalous traffic in the network for the topology shown in Fig. 5. The network was tested in static scenario and also in a scenario in which nodes are mobile. XAI technique has to be trained with traffic from normal data as well as anomalous data. To generate the data, node 9 was elected as the sinkhole attacker and node 8 was selected as the blackhole attacker. The simulation scenario is summarized in Table 9. We have introduced the mobility at 30 s after the start of simulation. The attacks were introduced at 120 s after the start of the simulation. From Figs. 20 and 21, it can be observed that a PDR of 100% was obtained before the introduction of mobility at 30 s. However, with the introduction of mobility, slight drop in PDR can be observed initially. PDR drop is more severe in the case of sinkhole attack than blackhole attack because sinkhole attacker attracts more nodes to send the traffic towards it and then drops the packets. As per the proposed technique, the attacker node is eliminated from the parent list if it is detected as anomalous for a duration of 3 time windows. Due to the novel parent discovery in RPL*, process the PDR increases after 210 s for both sinkhole and blackhole attacks even though attacker nodes are still present in the network. The rules extracted for the above mentioned scenario are shown in Table 10. It

Table 10
Rules extracted for anomaly detection in RPL*

Rules	Description
1	if(NPC $\leq$ 1.62578256457382 and NPC $>$ 0.08280208896976061 and NC $>$ 1.7438704258500928 and UDP_forwarded $\leq$ 0.0361022330057352 and UDP_transmitted $\leq$ 2.33048079085093 and UDP_transmitted $>$ 1.6537382894445853 and UDP_received $>$ 11.621236156495863)
2	if(NPC $\leq$ 2.3377105632506825 and NPC $>$ 0.8041354309786741 and NC $\leq$ 3.64441842196715 and NC $>$ 1.375193700369072 and UDP_forwarded $\leq$ 1.51911860961426 and UDP_transmitted $\leq$ 2.4947873891772643 and UDP_received $>$ 16.4492106691232)
3	if(NPC $\leq$ 1.900157019449471 and NPC $>$ 0.8041354309786741 and UDP_forwarded $\leq$ 1.51911860961426 and UDP_received $\leq$ 2.7286315074789487 and UDP_received $>$ 1.6384444090716748)
4	if(NPC $\leq$ 1.62578256457382 and NPC $>$ 0.08280208896976061 and NC $>$ 1.7438704258500928 and UDP_forwarded $\leq$ 0.4360045 and UDP_transmitted $\leq$ 2.678245 and UDP_received $>$ 14.473445853)
5	if(NPC $>$ 0.3779905835995407 and NC $\leq$ 2.4432980215378564 and NC $>$ 1.656024932235491 and UDP_forwarded $\leq$ 0.23335775650728388 and UDP_transmitted $\leq$ 2.225259870395651 and UDP_transmitted $>$ 1.6245952922034057 and packet_forwarding $\leq$ 0.5873905395098917 and UDP_received $\leq$ 13.492289319863678)
6	if(NPC $\leq$ 1.62578256457382 and NPC $>$ 0.08280208896976061 and NC $>$ 0.2076476906461863 and UDP_forwarded $\leq$ 0.9118457375615117 and UDP_transmitted $\leq$ 2.505622550614934 and UDP_transmitted $>$ 1.6537382894445853 and UDP_received $\leq$ 3.7649316493258493 and Packet_forwarding $\leq$ 0.2202273314912931)
7	if(NPC $>$ 0.3779905835995407 and NC $\leq$ 1.656024932235491 and UDP_forwarded $\leq$ 0.23335775650728388 and UDP_transmitted $\leq$ 2.225259870395651 and UDP_transmitted $>$ 1.6245952922034057 and UDP_received $\leq$ 15.081086373925714 and UDP_received $>$ 1.333542624643791 and Packet_forwarding $\leq$ 0.2873905395098917)
8	if(NPC $>$ 0.3779905835995407 and NC $\leq$ 2.5750029657850986 and NC $>$ 1.656024932235491 and UDP_forwarded $\leq$ 0.23335775650728388 and UDP_transmitted $\leq$ 2.225259870395651 and UDP_transmitted $>$ 1.6245952922034057 and UDP_received $\leq$ 19.473162882250342 and UDP_received $>$ 13.492289319863678)
9	if(NPC $\leq$ 2.3377105632506825 and NPC $>$ 0.8041354309786741 and NC $>$ 1.375193700369072 and UDP_transmitted $\leq$ 1.51911860961426 and UDP_transmitted $\leq$ 2.4947873891772643 and UDP_received $\leq$ 16.4492106691232 and UDP_received $>$ 3.8848541946545496)
10	if(NPC $>$ 0.3779905835995407 and NC $>$ 2.5750029657850986 and UDP_forwarded $\leq$ 0.23335775650728388 and UDP_transmitted $\leq$ 2.225259870395651 and UDP_transmitted $>$ 1.6245952922034057 and UDP_received $\leq$ 19.473162882250342 and UDP_received $>$ 13.492289319863678 and Packet_forwarding $\leq$ 0.2873905395098917)

Table 11
Performance of RPL* for various attack scenarios.

Scenario	Description	Accuracy	Precision	Recall	F1-score
1	Node 9 is sinkhole attacker and node 8 is blackhole attacker	0.98	0.98	0.98	0.98
2	Node 2 sinkhole attacker and node 8 blackhole attacker	0.99	0.99	0.95	0.97
3	Node 3 sinkhole attacker and node 8 blackhole attacker	1.00	1.00	1.00	1.00
4	Node 6 sinkhole attacker and node 8 blackhole attacker	0.98	0.99	0.93	0.96

may be noted that the same rules were used to isolate other anomalous situations as mentioned in Table 11 and have obtained a F1-score of over 95% for all scenarios mentioned in Table 11. The receiver operating characteristic (ROC) curve corresponding to the scenarios mentioned in Table 11 are presented in Fig. 22. This indicates that the proposed unsupervised technique can effectively isolate anomalous scenarios for which it is not trained for.

6. Conclusion

In this work, we have proposed RPL* - an XAI-based routing protocol for proactive routing in IoMT applications. The proposed protocol is robust to handling mobility in nodes and is resilient to routing attacks when compared to the widely used RPL protocol. The modification to support the mobility in RPL* is achieved via three components — priority setting, parent selection and connectivity management. The proposed mobility detection mechanism in RPL* will identify the mobility in the network in an energy efficient manner without incurring additional communication overhead. Any deviation from the normal behaviors of the

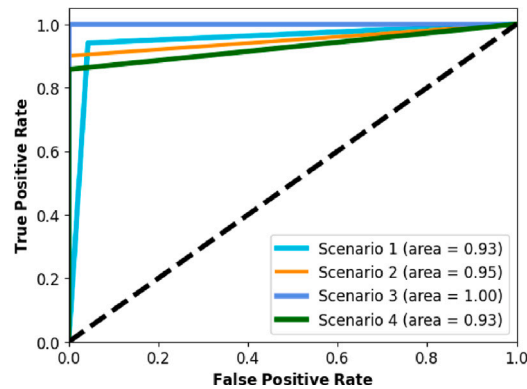


Fig. 22. ROC curves corresponding to anomaly classification.

network are identified as anomalies using an unsupervised Explainable Artificial Intelligence (XAI) strategy. The carefully designed novel proactive mobility management mechanism ensures that the communication overhead is minimized through the transmission of the DIS and DIO by only relevant nodes. This mechanism facilitates seamless parent transitions when necessary, preventing network partitioning due to mobility. In our evaluations, RPL* achieved an F1-score exceeding 0.96 for detecting black hole and sinkhole attacks in IoMT test scenarios. Furthermore, we assessed the performance of RPL* across various speeds, varying numbers of mobile nodes, and different network topologies. Additionally, RPL* was tested using diverse mobility models, including random, structured, and group mobility, catering to a wide range of applications. The results indicate that RPL* is a potential alternative to RPL protocol. Although the codes developed in Cooja for the sky motes can be run on the sky motes as it is, testing of the protocol using sky motes as well as other IoMT hardware will be considered in our future work.

CRedit authorship contribution statement

Sumitra Budania: Visualization, Methodology, Investigation, Data curation. **Meetha V. Shenoy:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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